

Database Design & normalization

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Why?

- Why ? Why ? Why?
- Why we need to talk about database design?

- Actually there is a very deep bounding between OO and Database

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Some History about DataBase

- There is a time about how to store a database
- Several data structures are used and implemented in commercial products
 - Tree (pointer)
 - Graph (pointer)
 - Linked list of ...
 -
 - Hetrogenious

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History

Following the technology progress in the areas of processors, computer memory, computer storage, and computer networks, the sizes, capabilities, and performance of databases and their respective DBMSs have grown in orders of magnitude. The development of database technology can be divided into three eras based on data model or structure: navigational,^[9] SQL/relational, and post-relational.

The two main early navigational data models were the hierarchical model, epitomized by IBM's IMS system, and the CODASYL model (network model), implemented in a number of products such as IDMS.

The relational model, first proposed in 1970 by Edgar F. Codd, departed from this tradition by insisting that applications should search for data by content, rather than by following links. The relational model employs sets of ledger-style tables, each used for a different type of entity. Only in the mid-1980s did computing hardware become powerful enough to allow the wide deployment of relational systems (DBMSs plus applications). By the early 1990s, however, relational systems dominated in all large-scale data processing applications, and as of 2015 they remain dominant:

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The Father of relational database

Edgar F. Codd

From Wikipedia, the free encyclopedia
(Redirected from E. F. Codd)

Edgar Frank "Ted" Codd (19 August 1923 – 18 April 2003) was an English computer scientist who, while working for IBM, invented the relational model for database management, the theoretical basis for relational databases and relational database management systems. He made other valuable contributions to computer science, but the relational model, a very influential general theory of data management, remains his most mentioned, analyzed and celebrated achievement.^{[6][7]}

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1 Biography
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Edgar "Ted" Codd



Born Edgar Frank Codd
19 August 1923^[18]
Farnham, Dorset, England

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Concepts of Relational Database

Customer						Table, Relation	
cust_id	lastname	firstname	address	postal_code			
1	Cramer	John	213 Main St.	22160			
2	Adams	Steven	333 Bering St.	33140			
3	Cramer	Ann	14 Wadhurst Rd.	50329			
4	Martin	Andrew	744 Baker Blvd.	22200			
5	Smith	Patricia	55 Jeffer Way	52100			
6	Piggs	Robert	62 Polk St.	50920			
7	Mandv	Malen	771 Ina St	22700			

Rent						Row, Record, Tuple	
rent_id	cust_id	reg_no	rent_date	return_date			
1	ACC-223	29.12.2004	4.1.2005				
2	BAA-224	2.1.2005	5.1.2005				
3	BAA-441	6.1.2005	8.1.2005				
4	ABC-122	11.1.2005	17.1.2005				
5	CCE-326	15.1.2005	17.1.2005				
6	ACC-223	19.1.2005	20.1.2005				
7	BAA-441	22.1.2005	26.1.2005				
8	ABC-122	26.1.2005					
9	BAA-441	26.1.2005					
10	BAA-441	26.1.2005					

Car Type				Car			Field	
model_id	mark	model	year	reg_no	model_id		Column, Attribute	
1	Ford	Focus	2004	ABC-112	11			
2	Ford	Mondeo	2005	ABC-122	11			
3	Peugeot	307	2004	ABC-123	11			
4	Peugeot	407	2005	ACC-223	11			
5	Renault	Clio	2004	ACC-224	11			
6	Renault	Laguna	2003	ACC-667	11			
				BAA-441	11			
				BAA-442	11			
				BAA-224	11			
				CCE-326	11			

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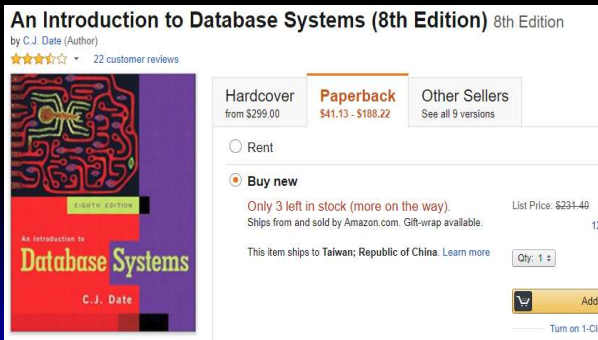
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If you really want to know deeper about normalization



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Writing a DB application

- Well, it is considered an easy topic in CS but mostly emphasized in MIS
- If you know DBMS programming, it won't make you get paid higher

- Why should we care?

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Let's start with an example.

Say you need a sales report something like this:

Customer			Catalog		Unit Price	Date	Qty Sold	Actual Price	Extended Price
No.	Name	Address	No.	Description					
131	Jo Blo	13 May St	3A21	T-Shirt	12.49	03/01/98	45	10.00	450.00
179	Yo Yo	271 OK Ave	1B77	Sweats	15.00	01/03/98	12	15.00	180.00
212	Mu Mu	32 Saddle Rd	4X21	Pants	23.47	12/11/98	5	21.00	105.00
.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.

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What the *uninitiated* (read "*amateur*") database designer tends to do

is to build a **relational table** that *mimics this report*.

- That is, it has the **same columns** as this report.
- But what would we call this class?
- The best name would probably be something like "**Sales**" or "**Sales Analysis.**"

But . . .

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The problem is that we have **three kinds of data in this report**.

- We have:

Data that describes a **Customer** (Cust No./Name/Address)

Data that describes a **Product**
(Cat No./Description/Unit Price)

And data that describes a **Sale** (Date/Quantity/Actual and Extended Prices)

- Compare this situation with all the earlier models we have looked at
- You'll see that **Customer**, **Product** and **Sale** should each be a *separate class . . .*

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The maintenance horror of the poorly designed database

- A customer can continuously buy several kinds of product.
- What if he change his name?
- What if the price of a product is increased or decreased?
- What if a customer change its address?

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What is the problem of the amateur's database design?

- This structure **does not** allow our database to answer

any query

that could possibly be dreamed up against that data.

- Some query can be done but very inefficient

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- The “Un-normalized” structure that mimicked the report will have problems , down the line a few months or years,
- Attempting to answer queries that the database designer did not foresee -
- What I refer to as:

“That most dreaded of all database phenomena,

Unanticipated Queries”

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Normalization

What Normalization is for is to make sure that each database table carries only the attributes that **actually describe** What is needed.

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Normalization

- Definition: Normalization is the process of structuring relational database schema such that most ambiguity is removed. The stages of normalization are referred to as normal forms and progress from the least restrictive (First Normal Form) through the most restrictive (Fifth Normal Form). Generally, most database designers do not attempt to implement anything higher than Third Normal Form or Boyce-Codd Normal Form.

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A simpler explanation to normalization

There are two goals of the normalization process:

- **eliminate redundant data** (for example, storing the same data in more than one table) and
- **ensure data dependencies** make sense (only storing related data in a table). Both of these are worthy goals as they reduce the amount of space a database consumes and ensure that data is logically stored.

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Normal forms

- The database community has developed a series of guidelines for ensuring that databases are normalized. These are referred to as normal forms and are numbered from one (the lowest form of normalization, referred to as first normal form or **1NF**) through five (fifth normal form or **5NF**).
- In practical applications, you'll often see **1NF**, **2NF**, and **3NF** along with the occasional 4NF. Fifth normal form is very rarely seen and won't be discussed in this article.

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Normal form hierarchy

- **First normal form (1NF)** sets the very basic rules for an organized database:
 - Eliminate duplicative columns from the same table.
 - Create separate tables for each group of related data and identify each row with a unique column or set of columns (the primary key).
- **Second normal form (2NF)** further addresses the concept of removing duplicative data:
 - Meet all the requirements of the first normal form.
 - Remove subsets of data that apply to multiple rows of a table and place them in separate tables.
 - Create relationships between these new tables and their predecessors through the use of foreign keys.
- **Third normal form (3NF)** goes one large step further:
 - Meet all the requirements of the second normal form.
 - Remove columns that are not dependent upon the primary key.
- **Finally, fourth normal form (4NF)** has one additional requirement:
 - Meet all the requirements of the third normal form.
 - A relation is in 4NF if it has no multi-valued dependencies.

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1ST NF

- Eliminate duplicative columns from the same table.
- Create separate tables for each group of related data and identify each row with a unique column or set of columns (the primary key).

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An classic example

- a table within a human resources database that stores the manager-subordinate relationship.
- For the purposes of our example, we impose the business rule that each **manager** may have one or more **subordinates** while each subordinate may have only one manager.

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An intuitive table

Manager	Subordinate1	Subordinate2	Subordinate3	Subordinate4
Bob	Jim	Mary	Beth	
Mary	Mike	Jason	Carol	Mark
Jim	Alan			

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Why it is not even 1st NF?

- recall the first rule imposed by 1NF: eliminate duplicative columns from the same table.? Clearly, the Subordinate1-Subordinate4 columns are duplicative.
- Jim only has one subordinate, the Subordinate2-Subordinate4 columns are simply wasted storage space
- Furthermore, Mary already has 4 subordinates ?what happens if she takes on another employee? The whole table structure would require modification.

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A second bright idea

- Let try something like this:

Manager	Subordinates
Bob	Jim, Mary, Beth
Mary	Mike, Jason, Carol, Mark
Jim	Alan

- **Manager Subordinates** Bob Jim, Mary, Beth Mary Mike, Jason, Carol, Mark Jim Alan This solution is closer, but it also falls short of the mark
- The subordinates column is still duplicative and non-atomic. What happens when we need to add or remove a subordinate?? We need to read and write the entire contents of the table.? That not a big deal in this situation, but what if one manager had one hundred employees??Also, it complicates the process of selecting data from the database in future queries.

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Here is a table that satisfies the first rule of 1NF:

Manager	Subordinate
Bob	Jim
Bob	Mary
Bob	Beth
Mary	Mike
Mary	Jason
Mary	Carol
Mary	Mark
Jim	Alan

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Not finished yet

- Now, what about the second rule: identify each row with a unique column or set of columns (the primary key)
- You might take a look at the table above and suggest the use of the subordinate column as a primary key. In fact, the subordinate column is a good candidate for a primary key due to the fact that our business rules specified that each subordinate may have only one manager.
- However, the data that we have chosen to store in our table makes this a less than ideal solution.? What happens if we hire another employee named Jim? How do we store his manager-subordinate relationship in the database??

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Finally, the 1st normal form

Manager	Subordinate
182	143
182	201
182	123
201	156
201	041
201	187
201	196
143	202

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Towards to 2NF

- Definition:** In order to be in Second Normal Form, a relation must first fulfill the requirements to be in First Normal Form. Additionally, each nonkey attribute in the relation must be functionally dependent upon the primary key.

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An example

Order #	Customer	Contact Person	Total
1	Acme Widgets	John Doe	\$134.23
2	ABC Corporation	Fred Flintstone	\$521.24
3	Acme Widgets	John Doe	\$1042.42
4	Acme Widgets	John Doe	\$928.53

The relation is in First Normal Form, but not Second Normal Form:

Remove subsets of data that apply to multiple rows of a table and place them in separate tables

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Two tables to satisfy 2NF

Customer	Contact Person
Acme Widgets	John Doe
ABC Corporation	Fred Flintstone

Order #	Customer	Total
1	Acme Widgets	\$134.23
2	ABC Corporation	\$521.24
3	Acme Widgets	\$1042.42
4	Acme Widgets	\$928.53

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comments

- The creation of two separate tables eliminates the dependency problem experienced in the previous case.
- In the first table, contact person is dependent upon the primary key -- customer name. The second table only includes the information unique to each order.
- Someone interested in the contact person for each order could obtain this information by performing a JOIN operation.

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3RD NF

- **Definition:** In order to be in Third Normal Form, a relation must first fulfill the requirements to be in Second Normal Form. Additionally, Every non-prime attribute of R is non-transitively dependent (i.e. directly dependent) on every candidate key of R.
- **A short explanation: No Dependencies on Non-Key Attributes**

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- In the relational model of databases, a **candidate key** of a relation is a minimal superkey for that relation; that is, a set of attributes such that
- the relation does not have two distinct tuples (i.e. rows or records in common database language) with the same values for these attributes (which means that the set of attributes is a superkey)
- there is no proper subset of these attributes for which (1) holds (which means that the set is minimal).
- The constituent attributes are called **prime attributes**. Conversely, an attribute that does not occur in ANY candidate key is called a **non-prime attribute**.

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An example

Company	City	State	ZIP
Acme Widgets	New York	NY	10169
ABC Corporation	Miami	FL	33196
XYZ, Inc.	Columbia	MD	21046

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The answer

- In this example, the city and state are dependent upon the ZIP code. To place this table in 3NF, two separate tables would be created -- one containing the company name and ZIP code and the other containing city, state, ZIP code pairings.

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To go or not to go higher?

- This may seem overly complex for daily applications and indeed it may be. Database designers should always keep in mind the tradeoffs between higher level normal forms and the resource issues that complexity creates.
- For example, in the last slide, if the city name and state name shall never change in 1000 years, the only harm is the duplication of data. The impact of change is of 0 possibility.

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An exercise

(20分) 假設你負責分析一個系統，此系統的資料包含了下面許多欄位
Please analyze a system which contains the following attributes

S#: 零件供應商的編號(Supplier no)
SNAME: 零件供應商的姓名(supplier name)
CITY1 零件供應商的城市(The city of a supplier)
P# 零件編號 (part no.)
PNAME 零件名稱 (part name)
COLOR 零件色彩 (part color)
WEIGHT 零件重量 (part weight)
CITY2 零件所儲存的城市 (city where the parts are stored)
QTY 零件的存量 (The quantity of the parts)

In your analysis, you found that a part can be supplied by several suppliers. Please determine how many tables should be used and what is the content of each table.

在你分析的過程中，你發現一個零件可能有多個供應商可以供應。請簡單說明這樣一個資料庫系統，你要用幾個表格，每個表格的屬性又為何？

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